

STAC67: Regression Analysis

Lecture 6

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Review of Lecture 5

Test for the slope. $H_0 : \beta_1 = \beta_1^0$ vs $H_a : \beta_1 \neq \beta_1^0$ (or one-sided version, with $<$ or $>$)

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Not normal, because σ is **estimated** rather than known:

$$\hat{\sigma} = \sqrt{MSE} = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{\sum_{i=1}^n e_i^2}{n-2}}, \quad SE(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{S_{xx}}}$$

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100(1 - α)% **confidence interval** for β_1 : estimate $\pm$ margin of error

$$\hat{\beta}_1 \pm \underbrace{t(1 - \alpha/2; n - 2)}_{\text{student quantile}} \cdot \underbrace{\frac{\hat{\sigma}}{\sqrt{S_{xx}}}}_{SE(\hat{\beta}_1)}$$

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Estimate β_1 with a 99 % confidence interval. Interpret the interval estimate.

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$$\begin{aligned}\hat{\beta}_1 \pm t(0.995; 82) SE(\hat{\beta}_1) &= -170.575 \pm 2.637 \times 41.574 \\ &= (-280.21, -60.94)\end{aligned}$$

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- **Interpretation** With 99% confidence, the mean crime rate falls by between 60.9 and 280.2 offences per 100,000 for each additional percentage point of high-school graduates.

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- **Interpretation** With 99% confidence, the mean crime rate falls by between 60.9 and 280.2 offences per 100,000 for each additional percentage point of high-school graduates.
- Zero is **not** in the interval, so at the 1% level there is evidence of a linear relationship between X and Y .

In R: `confint(fit, level = 0.99)`.

Interval Estimation of $E(Y|X = x_0)$

Suppose that x_{n+1} is a new value of x for which we want to do prediction.

1. Estimation of $\mu_{n+1} = E[Y|X = x_{n+1}]$
2. Prediction of Y value for an individual with $X = x_{n+1}$
 - We use fitted regression model to do both of these
 - Estimation of

$$\mu_{n+1} = \beta_0 + \beta_1 X_{n+1}$$

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- We estimate it by the fitted value at x_{n+1} :

$$\hat{Y}_{n+1} = \hat{\beta}_0 + \hat{\beta}_1 x_{n+1}, \quad E(\hat{Y}_{n+1}) = \beta_0 + \beta_1 x_{n+1} = \mu_{n+1} \text{ (unbiased)}$$

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Let's now derive the variance formula as well.

Derivation of $Var(\hat{Y}_{n+1})$

Recall from Lecture 4, the variances of the estimators:

$$Var(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right), \quad Var(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}$$

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Re-centre the fitted value, using $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$:

$$\hat{Y}_{n+1} = \hat{\beta}_0 + \hat{\beta}_1 x_{n+1} = \bar{Y} + \hat{\beta}_1 (x_{n+1} - \bar{x})$$

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Also from Lecture 4: We have two terms that are uncorrelated $Cov(\bar{Y}, \hat{\beta}_1) = 0$, so the variances add:

$$Var(\hat{Y}_{n+1}) = Var(\bar{Y}) + (x_{n+1} - \bar{x})^2 Var(\hat{\beta}_1) = \frac{\sigma^2}{n} + (x_{n+1} - \bar{x})^2 \frac{\sigma^2}{S_{xx}}$$

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$$Var(\hat{Y}_{n+1}) = \sigma^2 \left[\frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}} \right]$$

$$SE(\hat{Y}_0) = \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}}}$$

Derivation of $Var(\hat{Y}_{n+1})$

Analysis

- It is the smallest at $x_{n+1} = \bar{x}$, and growing as you move away: the line is best determined *in the middle of the data*.
- That is why, as we will see below, the confidence band is pinched at the centre and flares at the ends.
- Also, it decreases as n or the variability of X grows.

Sanity check: at $x_{n+1} = 0$, $\hat{Y}_{n+1} = \hat{\beta}_0$, and the box gives back $Var(\hat{\beta}_0)$ above.

Confidence Interval for $\mu_{n+1} = E(Y|X = x_{n+1})$

- A $(1 - \alpha)\%$ confidence interval for μ_{n+1} :

$$\hat{Y}_{n+1} \pm t(1 - \alpha/2; n - 2)SE(\hat{Y}_{n+1})$$

- Exercise: Obtain 95% confidence interval for the mean crime rate for states of high school graduate rate of 80%.

```
new.data = data.frame(X=80)
predict(fit, new.data, interval="confidence")
```

```
##           fit          lwr          upr
## 1 6871.585 6347.116 7396.054
```

Prediction of new observation

The value of Y for an individual with $X = x_{n+1}$ is:

$$Y_{n+1} = \beta_0 + \beta_1 X + \epsilon_0$$

- **Estimate of Y_{n+1} :** $\hat{Y}_{n+1} = \hat{\beta}_0 + \hat{\beta}_1 X_{n+1} = \hat{\mu}_0$
- **Prediction error:** $e_{n+1} = Y_{n+1} - \hat{Y}_{n+1}$
- $Var(e_{n+1}) = Var(Y_{n+1} - \hat{Y}_{n+1}) = Var(Y_{n+1}) + Var(\hat{Y}_{n+1})$, since the new observation (x_{n+1}, Y_{n+1}) is independent of the sample the line was fitted on

$$= \sigma^2 + \sigma^2 \left[\frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}} \right] = \sigma^2 \left[\mathbf{1} + \frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}} \right]$$

$$s_{\{pred\}} = \hat{\sigma} \sqrt{\mathbf{1} + \frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}}}$$

The extra 1 is the noise of the new observation itself. It never shrinks, so a prediction interval stays wide no matter how much data you collect.

- $100 \times (1 - \alpha)\%$ **prediction Interval:**

$$\hat{Y}_{n+1} \pm t(1 - \alpha/2; n - 2) s_{\{pred\}}$$

Exercise

- Exercise: Obtain 95% prediction interval for the crime rate for states of high school graduate rate of 80%.

```
new.data = data.frame(X=80)
predict(fit, new.data, interval="prediction")
```

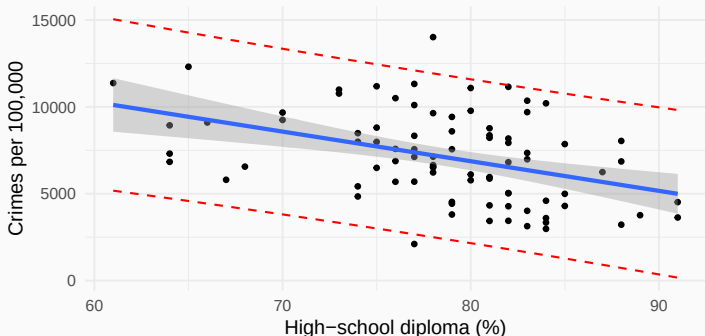
```
##           fit      lwr      upr
## 1 6871.585 2154.92 11588.25
```

Graph Prediction Intervals (using ggplot)

```
library(ggplot2)
```

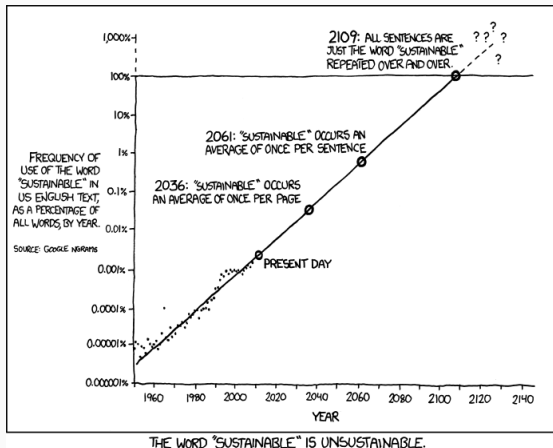
```
Crime.Pred <- predict(fit, interval = "prediction")  
new.df <- cbind(Crime, Crime.Pred)
```

```
ggplot(new.df, aes(X, Y)) +  
  geom_point(size = 0.9) +  
  geom_line(aes(y = lwr), colour = "red", linetype = "dashed") +  
  geom_line(aes(y = upr), colour = "red", linetype = "dashed") +  
  geom_smooth(method = lm, se = TRUE) +  
  labs(x = "High-school diploma (%)", y = "Crimes per 100,000") +  
  theme_minimal(base_size = 10)
```



Grey band: 95% CI for the mean response. Red dashed: 95% prediction interval.

How far can a line be pushed?



A line fitted to sixty years of data, extended a century past them. The $(x_0 - \bar{x})^2$ term widens our intervals away from the data, but only on the assumption that the line still holds out there. No interval protects against the model itself breaking.

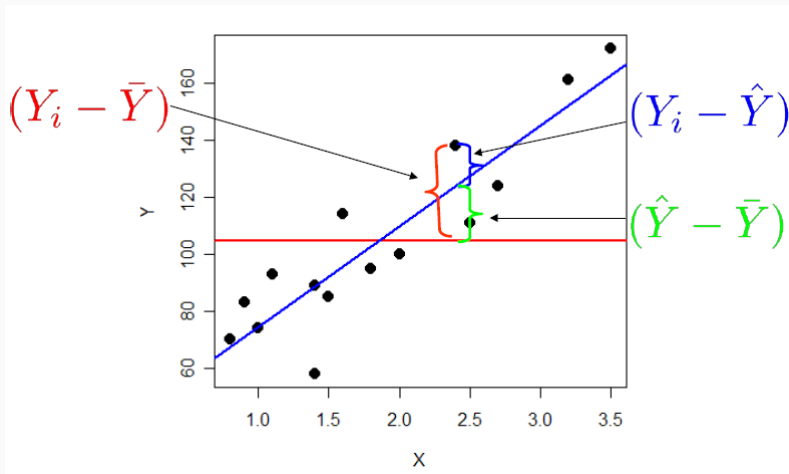
Analysis of Variance Approach to Regression Analysis

- How well does the least squares fit explain variation in Y ?

$$\begin{aligned}\sum_{i=1}^n (Y_i - \bar{Y})^2 &= \sum_{i=1}^n (Y_i - \hat{Y}_i + \hat{Y}_i - \bar{Y})^2 \\ &= \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 + \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2\end{aligned}$$

- Total Sum of Squares (SST):
- Model Sum of Squares (SSR): Variation in Y explained by the regression.
- Error Sum of Squares (SSE): Variation in Y that is left explained.

How does that breakdown look on a scatterplot?



Analysis of Variance Table

Source of Variation	Sum of Squares	df	Mean Squares
Regression	SSR	1	$MSR = SSR/1$
Error	SSE	$n - 2$	$MSE = SSE/(n - 2)$
Total	SST	$n - 1$	

- The total degrees of freedom is always $n - 1$, as we fit the mean only.
- In the simple regression, the degrees of freedom used by the model is $n - 2$: intercept and slope are fitted.
- F -test for $H_0 : \beta_1 = 0$

$$F = \frac{MSR}{MSE} = \frac{SSR/1}{SSE/n - 2}$$

- Under H_0 , $F \sim F(1, n - 2)$

- The F-distribution with k_1 and k_2 degrees of freedom can be defined as the distribution of the random variable F

$$F = \frac{V_1/k_1}{V_2/k_2},$$

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The same three ingredients as the t last week: two chi-squares instead of a normal and a chi-square.

- This is denoted as $F \sim F(k_1, k_2)$
- we can show that under $H_0 : \beta_1 = 0$,

$$\frac{MSR}{MSE} \sim F(1, n - 2)$$

Relationship b/w F-test and t-test

- We can rewrite SSR using the regression estimator:

$$SSR = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 = \sum_{i=1}^n \left(\hat{\beta}_1 (x_i - \bar{x}) \right)^2 = \hat{\beta}_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 = \hat{\beta}_1^2 S_{xx}$$

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using $\hat{Y}_i - \bar{Y} = \hat{\beta}_1 (x_i - \bar{x})$, the same re-centring as on the $Var(\hat{Y}_0)$ slide.

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- In the simple regression, this is equivalent to the test of

$$H_0 : \beta_1 = 0 \quad vs \quad H_1 : \beta_1 \neq 0$$

Descriptive Measure of Linear Association b/w X and Y

$$SST = SSR + SSE$$

$$1 = \frac{SSR}{SST} + \frac{SSE}{SST}$$

$\frac{SSR}{SST}$: the proportion of the total variation in Y explained by the regression

- A “good” model should have a large $R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$
- R^2 : Coefficient of determination

Coefficient of Determination

Properties

1. $0 \leq R^2 \leq 1$
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Indeed:

$$SST = \sum_{i=1}^n (Y_i - \bar{Y})^2 = S_{YY}, \quad SSR = \hat{\beta}_1^2 S_{xx} \text{ (slide 16)}$$

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and, from Lecture 3, $\hat{\beta}_1 = r\sqrt{S_{YY}/S_{xx}}$. Therefore

$$R^2 = \frac{SSR}{SST} = \frac{\hat{\beta}_1^2 S_{xx}}{S_{YY}} = r^2 \frac{S_{YY}}{S_{xx}} \cdot \frac{S_{xx}}{S_{YY}} = r^2$$

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Only in simple linear regression: we will see later that with several predictors, R^2 is not the square of a correlation.

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$$\frac{SSR}{\sigma^2} = \frac{\hat{\beta}_1^2 S_{xx}}{\sigma^2} = \left(\frac{\hat{\beta}_1 \sqrt{S_{xx}}}{\sigma} \right)^2 \sim \chi_{(1)}^2$$

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We already know $\frac{SSE}{\sigma^2} \sim \chi_{(n-2)}^2$, and SSR and SSE are independent (admitted).

So

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So

$$\frac{MSR}{MSE} = \frac{(SSR/\sigma^2)/1}{(SSE/\sigma^2)/(n-2)} \sim F(1, n-2)$$

Two independent chi-squares, each over its degrees of freedom.

Example (Crime Rate)

```
anova(fit)
```

```
## Analysis of Variance Table
##
## Response: Y
##           Df      Sum Sq  Mean Sq F value    Pr(>F)
## X             1  93462942  93462942   16.834 9.571e-05 ***
## Residuals    82  455273165   5552112
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- 1) Test whether or not there is a linear association between crime rate and percentage of high school graduates using an F test. Show the numerical equivalence of the two test statistics and decision rules.

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$$F^* = \frac{MSR}{MSE} = \frac{93,462,942}{5,552,112} = 16.83 \text{ on } (1, 82) \text{ df, } p = 0.0001 : \text{reject } H_0$$

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$$(t^*)^2 = (-4.103)^2 = 16.83 = F^* \quad \text{the two tests are one}$$

Example (Crime Rate), continued

2) Compute R^2 and r .

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$$R^2 = \frac{SSR}{SST} = \frac{93,462,942}{548,736,108} = 0.170, \quad r = -\sqrt{0.170} = -0.413$$

Example (Crime Rate), continued

2) Compute R^2 and r .

$$R^2 = \frac{SSR}{SST} = \frac{93,462,942}{548,736,108} = 0.170, \quad r = -\sqrt{0.170} = -0.413$$

 $R^2 = 0.17$ is small, yet the slope is significant at the 1% level.

- Significance answers *is there a relationship?*
- R^2 answers *how much of the variation does it explain?*
- They are different questions, and with $n = 84$ a weak relationship can still be clearly non-zero.